

Using VIX Regimes as Cross-Asset Trading Signals

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1 Overview

This report evaluates whether volatility regimes contain forward-looking predictive information for:

- directional equity exposure,
- volatility positioning,
- and broader cross-asset risk-factor implementation.

Using historical SPY and VIX data, volatility is segmented into structural market regimes and studied through the lens of:

- forward returns,
- realized volatility,
- signal persistence,
- and practical trade implementation.

Rather than treating VIX purely as a passive risk indicator, this framework evaluates whether volatility itself contains **tradable signal value**.

2 VIX Regime Definition

Regime	Definition
Low Vol	$VIX < 15$
Normal Vol	$15 \leq VIX < 20$
Elevated Vol	$20 \leq VIX < 30$
Stress Vol	$VIX \geq 30$

These regimes represent distinct market environments ranging from stable risk-taking periods to forced deleveraging and liquidity stress.

3 Market Context

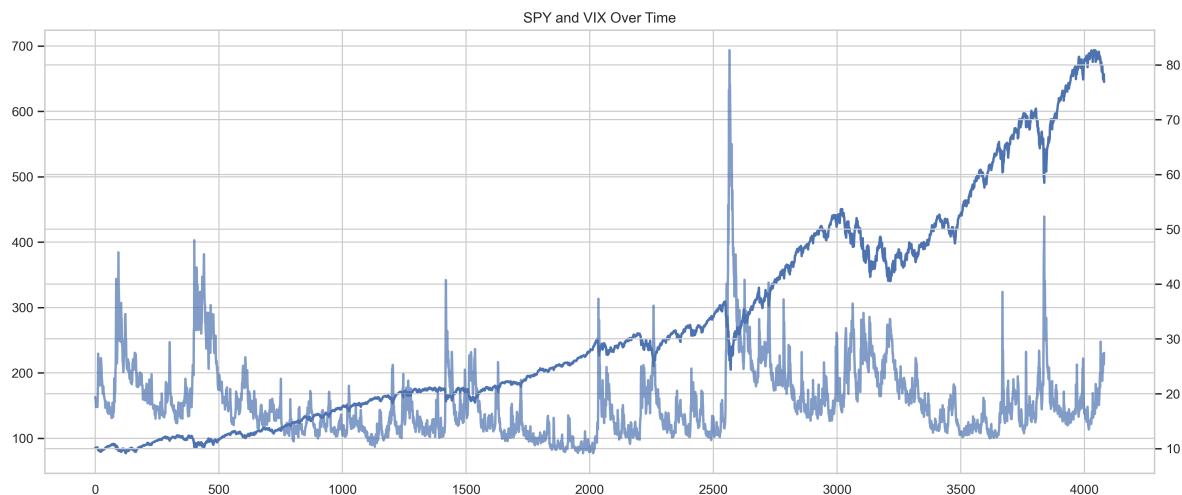


Figure 1: SPY and VIX Over Time

The inverse relationship between SPY and VIX is clear: volatility spikes coincide with sharp market drawdowns, while normalization phases often produce strong forward returns.

4 Sample Size by Regime

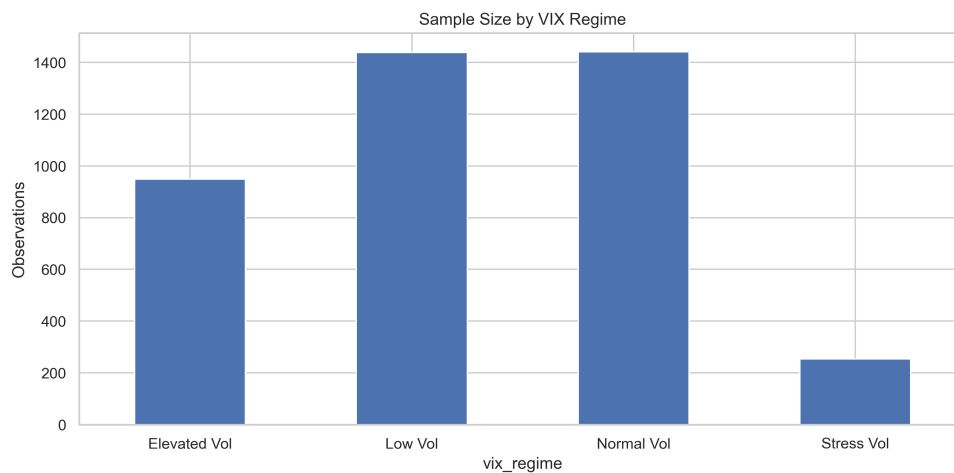


Figure 2: Observations by VIX Regime

Understanding sample size is critical when evaluating statistical significance. Stress regimes are rarer, but economically meaningful.

5 Forward Equity Returns

Table 1: Average Forward Returns by Regime

Regime	1D	5D	20D
Low Vol	0.000406	0.001432	0.006569
Normal Vol	0.000332	0.002072	0.008317
Elevated Vol	0.000675	0.004149	0.015760
Stress Vol	0.002474	0.010292	0.039222

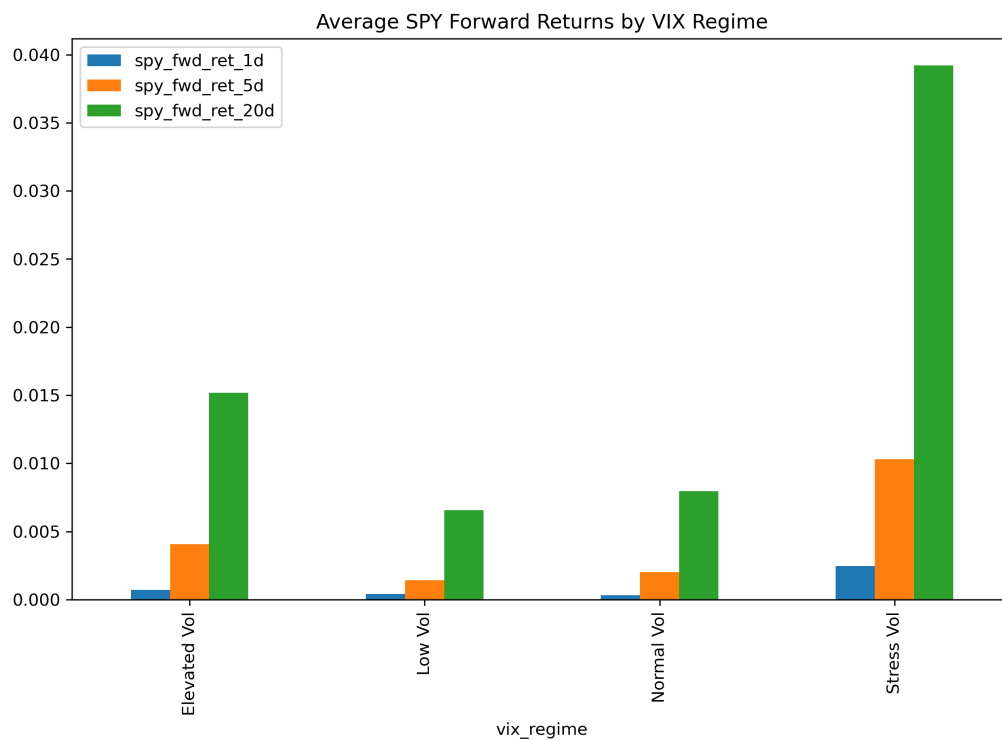


Figure 3: Average Forward Returns by Regime

Forward returns rise materially as volatility rises.

This likely reflects:

- panic-driven selling,
- liquidity dislocations,
- forced deleveraging,
- subsequent mean reversion.

Stress regimes produce the strongest forward return profile.

6 Signal Relationship

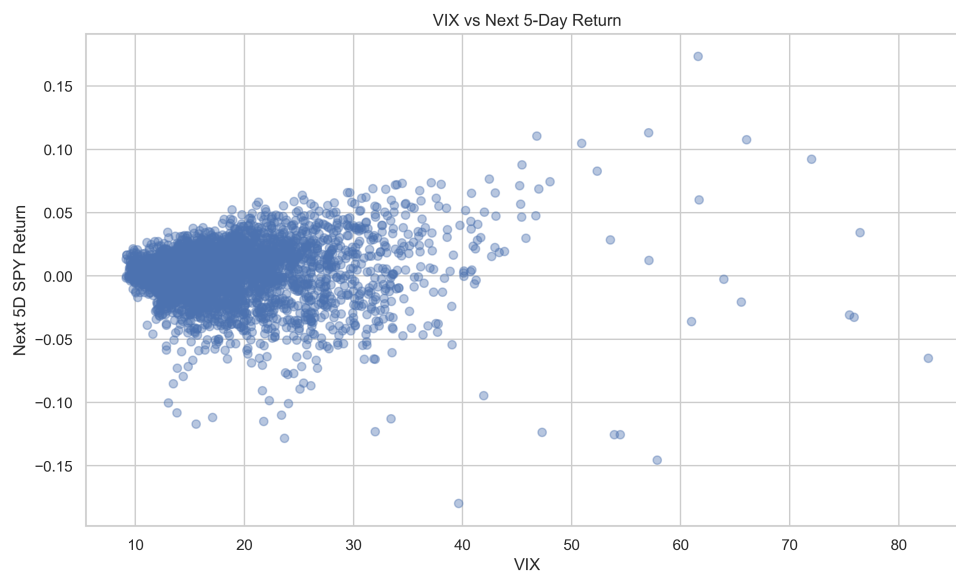


Figure 4: VIX vs Subsequent 5-Day SPY Return

While noisy at the single-observation level, elevated VIX levels skew toward stronger forward returns, supporting the broader regime-based signal.

7 Future Realized Volatility

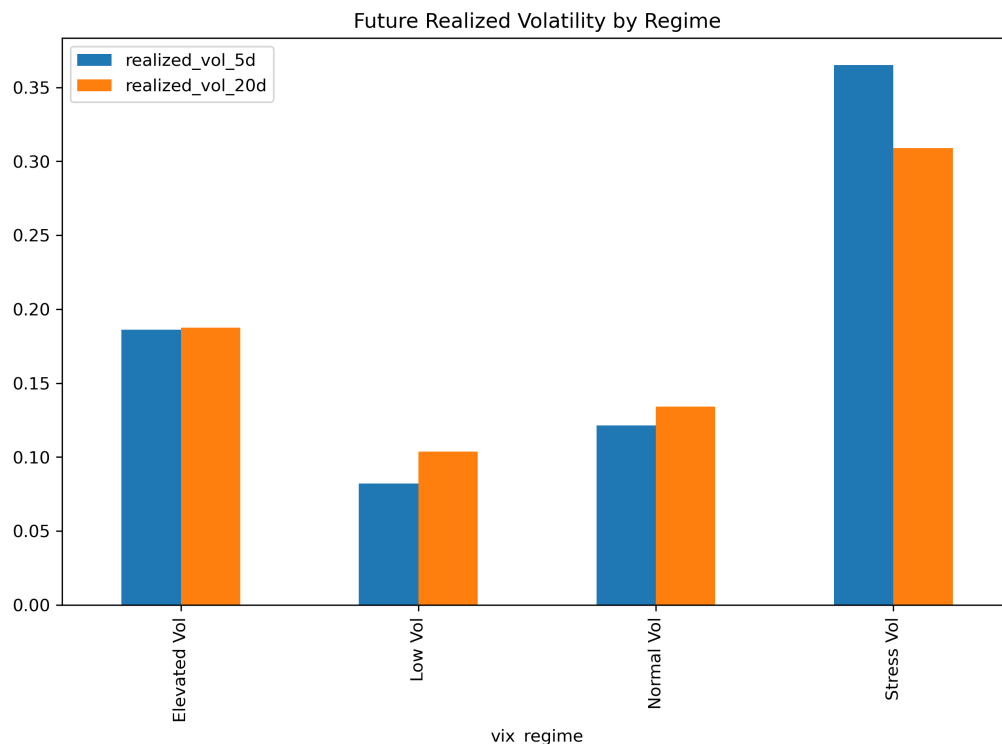


Figure 5: Future Realized Volatility by Regime

High VIX strongly predicts elevated future realized volatility.

This is important because:

$$\text{Volatility Edge} \approx \text{Realized Vol} - \text{Implied Vol}$$

If realized volatility exceeds implied volatility:

- long volatility,
- long gamma,
- convexity ownership

become attractive expressions.

If implied volatility exceeds realized volatility:

- short volatility,
- short gamma,
- premium harvesting

become attractive.

This directly connects regime analysis to options positioning.

8 Regime Persistence

Volatility clusters.

Once volatility enters elevated or stress regimes, it often remains elevated.

This persistence creates signal value because:

- elevated realized volatility persists,
- regime duration exceeds isolated event shocks,
- volatility positioning can be structured around persistence rather than noise.

This supports medium-horizon positioning.

9 Single-Day Spike Analysis

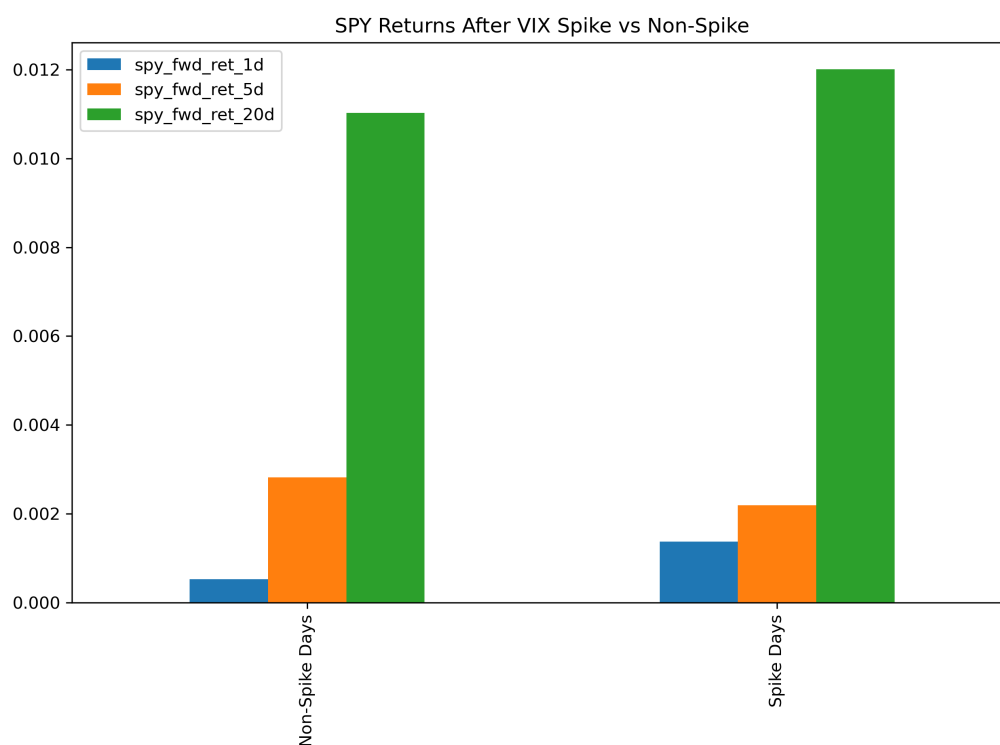


Figure 6: Forward Returns Following Large VIX Spikes

Single-day volatility spikes show weaker predictive power than full regime classification.

This suggests:

- isolated spikes are noisy,
- structural regime state matters more,
- persistence matters more than event shock.

Tradeable signals appear structural rather than event-driven.

10 Strategy Implementation

Simple implementation rules were tested:

- Long SPY when $VIX > 20$
- Long SPY when $VIX > 30$
- Long SPY when $VIX > 40$

Performance should be evaluated using:

- Sharpe ratio,
- Maximum Drawdown,
- Calmar ratio.

Raw Sharpe alone is incomplete. Large drawdowns may force position exit, making path-dependency critical.

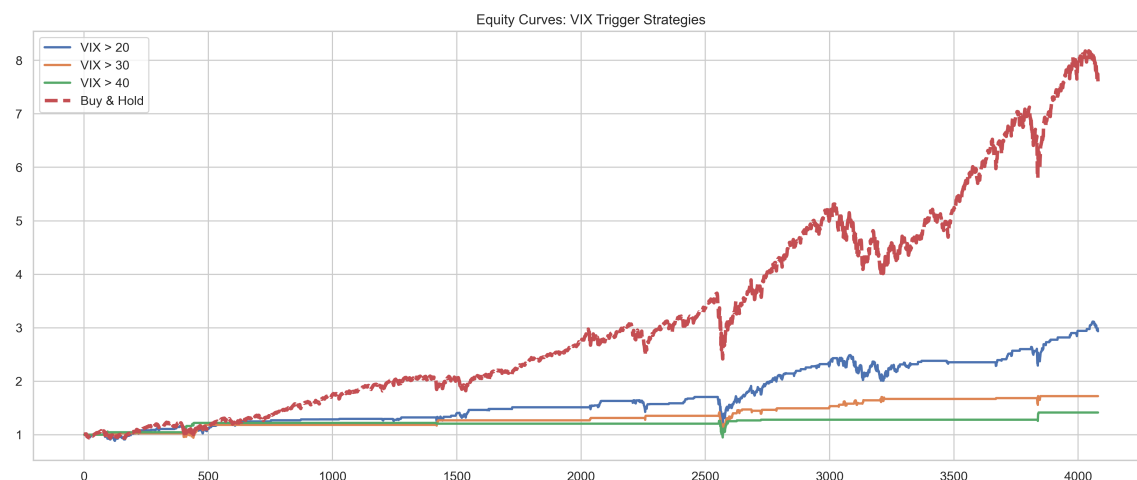


Figure 7: Equity Curves: VIX Trigger Strategies vs Buy-and-Hold

Implementation determines whether signal value survives in live trading.

11 Trade Expressions

Potential implementation includes:

- directional long equity exposure,
- long gamma positioning,
- volatility carry trades,
- skew positioning,
- dynamic hedge overlays.

Signal discovery is only the first step.
Trade construction determines realized alpha.

12 Conclusion

The strongest conclusion is not simply:

“buy stocks when VIX is high.”

That framing is incomplete.
A better conclusion is:

VIX regimes contain forward-looking predictive information that can be translated into directional equity, volatility, and risk-factor positioning.

High-volatility regimes are associated with:

- stronger forward returns,
- elevated realized volatility,
- persistent market states,
- clearer positioning opportunities.

The signal appears economically meaningful.
The next challenge is implementation, sizing, and risk control.